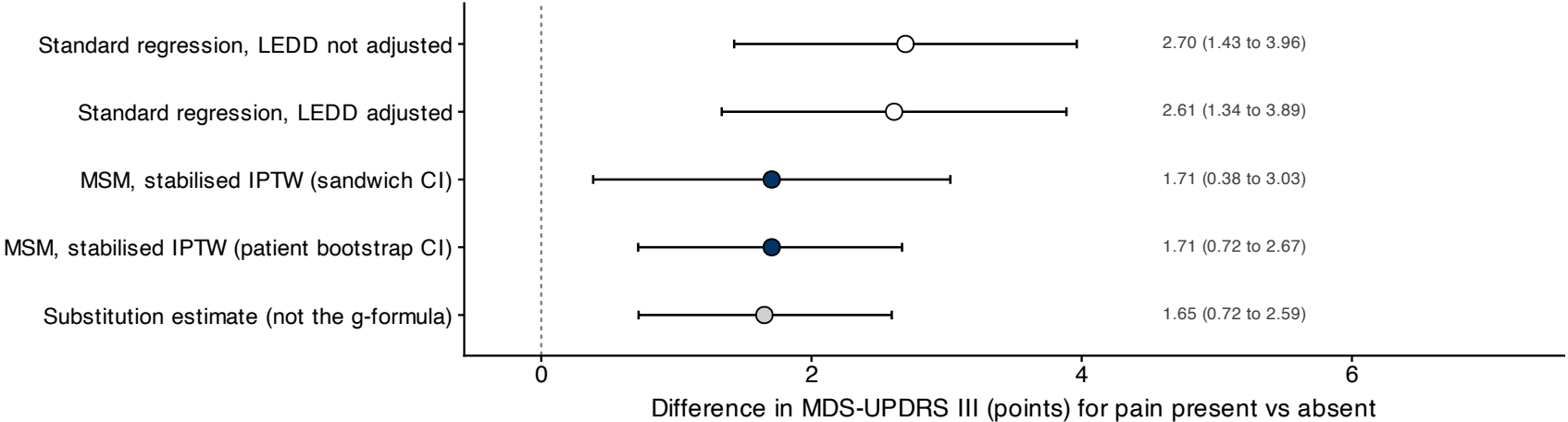


Figure 4. Handling dopaminergic dose as a time-varying confounder



Dopaminergic dose is affected by prior pain and in turn affects both later pain and motor severity, so conventional regression is biased whether or not it is adjusted for. Navy circles are the two causal estimates, which are about 37% smaller than the conventional ones. The grey circle holds the time-varying covariates at their observed values, which is not the g-formula, and is shown as a descriptive cross-check rather than as independent corroboration. All five models are fitted on the same 1,970 observations from 787 patients. LEDD, levodopa equivalent daily dose.